

ATPL EASA STUDY GUIDE | 090 - COMMUNICATIONS

Chapter 1: General Procedures & Phraseology

Syllabus Source: CAE Oxford (pp. 1-32) | Pass Target: >= 90% AviationExam

1. THE ICAO STANDARD RADIOTELEPHONY ALPHABET & NUMBERS

Aeronautical radiotelephony communications require absolute clarity and zero phonetic ambiguity. Standard ICAO phonetic spelling and number pronunciation must be used across all aeronautical frequencies:

Col 1	Col 2	Col 3	Col 4	Col 5	Col 6
A - Alpha	B - Bravo	C - Charlie	D - Delta	E - Echo	F - Foxtrot
G - Golf	H - Hotel	I - India	J - Juliett	K - Kilo	L - Lima
M - Mike	N - November	O - Oscar	P - Papa	Q - Quebec	R - Romeo
S - Sierra	T - Tango	U - Uniform	V - Victor	W - Whiskey	X - X-ray
Y - Yankee	Z - Zulu	0 - ZE-RO	1 - WUN	2 - TOO	3 - TREE
4 - FOW-er	5 - FIFE	6 - SIX	7 - SEV-en	8 - AIT	9 - NIN-er

[!] EXAM TRAP (AVIATIONEXAM) - AviationExam Number Transmission Rules (ICAO Doc 4444 / Annex 10)

- General Rule: Each digit must be transmitted individually (e.g. FL 180 is 'FLIGHT LEVEL WUN AIT ZE-RO'). - EXCEPTIONS for Whole Hundreds & Thousands: - 800 is transmitted as 'AIT HUND-red'. - 5,000 is transmitted as 'FIFE TOU-SAND'. - 11,000 is transmitted as 'WUN WUN TOU-SAND'. - 38,300 is transmitted as 'TREE AIT TOU-SAND TREE HUND-red'. - Altimeter Settings: 1013 hPa is 'WUN ZE-RO WUN TREE'. 29.92 inHg is 'TOO NIN-er DAY-SEE-MAL NIN-er TOO'. - 8.33 kHz Frequencies: All 6 digits must be spoken (e.g. 118.005 is 'WUN WUN AIT DAY-SEE-MAL ZE-RO ZE-RO FIFE'), EXCEPT when the 5th and 6th digits are both zero (e.g. 118.000 is 'WUN WUN AIT DAY-SEE-MAL ZE-RO').

2. READABILITY SCALE (SIGNAL STRENGTH & CLARITY 1 TO 5)

Scale Value	Official ICAO Definition	Operational Meaning & Quality
1	Unreadable	Signal cannot be understood at all.
2	Readable now and then	Only occasional words or syllables intelligible.
3	Readable but with difficulty	Message can be understood only with severe concentration and repeats.
4	Readable	Transmission clear with minor background noise or distortion.
5	Perfectly readable	Crystal clear audio quality; zero background noise or distortion.

3. STANDARD ICAO WORDS AND PHRASES

Phrase / Word	ICAO Definition	Operational Context & Exam Nuance
ROGER	I have received all of your last transmission.	CRITICAL: ROGER does NOT mean clearance, permission, or agreement! It means receipt only.

WILCO	I understand your message and will comply with it.	Combines ROGER and compliance. Never say 'ROGER WILCO' (redundant).
AFFIRM / NEGATIVE	'Yes' / 'No'	Standard affirmations. Never use conversational slang ('Yeah', 'Nope').
STANDBY	Wait and I will call you.	The caller must pause and wait; does not grant permission to act.
CORRECTION	An error has been made in this transmission...	The correct version follows immediately ('...correction heading 280').
CLEARED	Authorized to proceed under the conditions specified.	Reserved strictly for air traffic control clearances.

4. AIRCRAFT CALLSIGNS & ABBREVIATION RULES

- Type A (Registration): Characters corresponding to aircraft registration (e.g. G-ABCD). May be abbreviated to first letter and at least last two (G-CD).
- Type B (Operator + Registration): Telephony designator of operator + registration (e.g. SPEEDBIRD G-ABCD -> SPEEDBIRD CD).
- Type C (Operator + Flight Number): Telephony designator + flight number (e.g. IBERIA 1234). NEVER ABBREVIATED under any circumstances!
- Golden Rule of Abbreviation: An aircraft station may use its abbreviated callsign ONLY AFTER it has been addressed in this manner by an air traffic communication station!

[!] EXAM TRAP (AVIATIONEXAM) - Mandatory Readback Items (AviationExam Absolute Core - ICAO Annex 11)

The flight crew MUST read back verbatim: 1. ATC route clearances; 2. Clearances to enter, land on, take off on, hold short of, cross, taxi and backtrack on ANY runway; 3. Runway-in-use, altimeter settings (QNH, QFE), SSR codes (squawk), level instructions, heading and speed instructions; 4. Transition levels.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Algoritmo Paso a Paso (Guía para Dummies): Number Pronunciation & Readback

REGLAS MNEMOTÉCNICAS DE EXAMEN: 1. Pronunciación de números enteros: - Centenas redondas: 'HUND-red' (ej. 800 = 'AIT HUND-red'). - Millares redondos: 'TOU-SAND' (ej. 5.000 = 'FIFE TOU-SAND'). - Combinación: 38.300 = 'TREE AIT TOU-SAND TREE HUND-red'. 2. Niveles de vuelo (FL): SIEMPRE dígito a dígito (ej. FL 100 = 'FLIGHT LEVEL WUN ZE-RO ZE-RO'). 3. Escala de legibilidad: Del 1 (ininteligible) al 5 (perfecto). 'Read you 3' significa 'legible con dificultad'.

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: How should the frequency 118.005 MHz be transmitted over the radio under 8.33 kHz channel spacing?

- [A]: 'One eighteen point zero zero'
- [B]: 'One one eight decimal zero zero five'
- [C]: 'One one eight point zero five'
- [D]: 'One eighteen zero zero five'

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. In 8.33 kHz channel spacing, all six digits must be pronounced: 'One one eight decimal zero zero five'.

Q2: How should an altimeter setting of 1013 hPa be pronounced in ICAO radiotelephony?

- [A]: 'One thousand and thirteen'
- [B]: 'Ten thirteen'
- [C]: 'One zero one three'
- [D]: 'One zero thirteen'

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [C]. Altimeter settings are transmitted by pronouncing each digit separately: 'One zero one three'.

Q3: What does the ICAO standard word 'ROGER' mean when transmitted by an aircraft?

- [A]: I agree with your proposal
- [B]: I have received all of your last transmission
- [C]: I will comply with your instructions
- [D]: Cleared as requested

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. 'ROGER' means 'I have received all of your last transmission'. It does NOT mean compliance (which is 'WILCO') nor authorization.

Q4: When is a pilot permitted to use an abbreviated aircraft callsign (e.g. 'G-CD' instead of 'G-ABCD')?

- [A]: Anytime after initial contact
- [B]: Only after the ground station has first addressed the aircraft using the abbreviated callsign
- [C]: In emergency situations only
- [D]: When flying outside controlled airspace

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. A pilot must use full callsign until the ground station initiates the abbreviation; only then may the pilot use the abbreviated callsign.

Q5: Which of the following instructions must ALWAYS be read back verbatim by the pilot?

- [A]: Meteorological wind updates
- [B]: Altimeter setting (QNH), runway-in-use, and holding point instructions
- [C]: Time checks
- [D]: Surface temperature and dew point

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Runway instructions, holding points, QNH, squawks, headings, speeds, and levels are strictly mandatory readback items under ICAO Doc 4444.

Q6: How is a transmission readability of '3' defined on the ICAO 1 to 5 scale?

- [A]: Perfectly readable
- [B]: Readable with difficulty
- [C]: Readable now and then
- [D]: Unreadable

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. 1 = Unreadable, 2 = Readable now and then, 3 = Readable but with difficulty, 4 = Readable, 5 = Perfectly readable.

Q7: How should a pilot report climbing from FL 150 to FL 220 under standard ICAO radiotelephony?

- [A]: 'Climbing to two twenty'
- [B]: 'Leaving flight level one five zero, climbing to flight level two two zero'
- [C]: 'Up to flight level two two zero'
- [D]: 'Flight level two two zero now'

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. When vacating a level, both the vacated level and the cleared level must be stated: 'Leaving flight level one five zero, climbing to flight level two two zero'.

Q8: Which component of an air traffic message does NOT require mandatory verbatim readback by the flight crew?

- [A]: Altimeter setting (QNH)
- [B]: Surface wind and ambient temperature
- [C]: Transponder SSR squawk code
- [D]: Runway holding point instructions

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Surface wind, temperature, and runway surface condition are advisory reports and do not require verbatim readback, unlike instructions, levels, and QNH.

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Chapter 2: Propagation & Frequency Bands

Syllabus Source: CAE Oxford (pp. 33-60) | Pass Target: >= 90% AviationExam

1. THE RADIO FREQUENCY SPECTRUM & AERONAUTICAL BANDS

Band Name	Frequency Range	Aeronautical Applications
VLF (Very Low Frequency)	3 to 30 kHz	Worldwide navigation, submarine communications.
LF (Low Frequency)	30 to 300 kHz	Non-Directional Beacons (NDB), commercial broadcast.
MF (Medium Frequency)	300 to 3,000 kHz	NDBs, commercial AM radio (subject to night effect).
HF (High Frequency)	3 to 30 MHz	Long-range oceanic & polar aeronautical communications (Skywave).
VHF (Very High Frequency)	30 to 300 MHz	Aeronautical RTF (118.000 - 136.975 MHz), VOR (108.0-117.95 MHz), ILS Localizer (108.1-111.95 MHz), Emergency (121.500 MHz).
UHF (Ultra High Frequency)	300 to 3,000 MHz	ILS Glidepath (329.15-335 MHz), DME, SSR (1030/1090 MHz), GPS (1575.42 MHz), Military Guard (243.0 MHz).
SHF (Super High Frequency)	3 to 30 GHz	Airborne Weather Radar (9 GHz), MLS (5 GHz), Satellite communications.

2. VHF PROPAGATION & THEORETICAL RANGE FORMULA

VHF radio waves travel essentially by direct 'space wave' (line of sight). Atmospheric refraction bends waves slightly downward around the curvature of the Earth (effective earth radius = 4/3 of true radius):

[!] EXAM TRAP (AVIATIONEXAM) - The Master VHF Radio Range Formula (AviationExam Absolute Core)

Range (NM) = $1.23 \times (\sqrt{h_{\text{aircraft}} [\text{ft}]} + \sqrt{h_{\text{ground_station}} [\text{ft}]})$ -

Example 1: Aircraft at FL 100 (10,000 ft), ground station at sea level (0 ft): Range

= $1.23 \times \sqrt{10,000} = 1.23 \times 100 = 123 \text{ NM}$. - Example 2: Aircraft at FL 360 (36,000

ft), ground station at 400 ft: Range = $1.23 \times (\sqrt{36,000} + \sqrt{400}) = 1.23 \times$

$(189.7 + 20) = 1.23 \times 209.7 = \sim 258 \text{ NM!}$ - Note: Optical horizon uses 1.06; Radio

horizon uses 1.23 due to atmospheric refraction!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: VHF Line-of-Sight Range and 8.33 kHz Channeling

SCENARIO (AviationExam Standard Radiotelephony Calculation): An airliner is cruising at FL 360 (36,000 ft) en route to Madrid: - The Madrid ATC VHF transmitter antenna is located on a hill at 400 ft AMSL QUESTION: What is the maximum theoretical VHF two-way communication range in Nautical Miles (NM)? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Identify the Master Refracted Line-of-Sight Formula: - Master Equation: Range (NM) = $1.23 \times [\sqrt{h_{\text{aircraft}}} + \sqrt{h_{\text{station}}}]$ - (Factor is 1.23 for radio waves due to atmospheric refraction, vs 1.06 for visible light!). Step 2: Calculate the Square Roots of the Heights: - Aircraft Height $h_1 = 36,000 \text{ ft} \rightarrow \sqrt{36,000} = 189.74$ - Ground Station Height $h_2 = 400 \text{ ft} \rightarrow \sqrt{400} = 20.00$ - Sum of roots = $189.74 + 20.00 = 209.74$ Step 3: Multiply by the 1.23 Refraction Factor: - Range = $1.23 \times 209.74 = 257.98 \text{ NM}$ (~258 Nautical Miles)! FINAL ANSWER: Maximum VHF communication range is 258 NM! Beyond this distance, the direct VHF space wave is blocked by the curvature of the Earth.

4. HF SKYWAVE PROPAGATION & THE IONOSPHERE

High Frequency (HF: 3-30 MHz) propagates over thousands of miles via ionospheric refraction (skywave):

- Ionospheric Layers: D layer (50-90 km, daytime only, ABSORBS HF waves); E layer (90-140 km); F1 & F2 layers (140-400 km, merge into single F layer at night, REFLECTS HF waves back to Earth).
- Day vs Night Frequencies: DAY: Higher frequencies (10-18 MHz) required to penetrate the absorbing D layer and reach the F layer. NIGHT: Lower frequencies (3-8 MHz) required because the D layer vanishes and the F layer refracts lower frequencies.
- Dead Zone (Skip Zone): The area between the outer limit of the ground wave and the first returning sky wave where NO HF SIGNAL can be received.
- SELCAL (Selective Calling): An automated ground-to-air audio tone signaling system (4 letters, e.g. AB-CD). Alerts flight crew with chime and light, eliminating the need to listen to continuous static noise on HF.

4. 8.33 KHZ CHANNEL SPACING TRANSITION

To relieve severe VHF channel congestion in European airspace, the channel spacing was reduced from 25 kHz to 8.33 kHz (dividing each 25 kHz channel into three 8.33 kHz channels). Carriage of 8.33 kHz capable radio equipment is mandatory across all controlled airspace in the European ICAO region.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Algoritmo Paso a Paso (Guía para Dummies): Number Pronunciation & Reading**

REGLAS MNEMOTÉCNICAS DE EXAMEN: 1. Pronunciación de números enteros: - Centenas redondas: 'HUND-red' (ej. 800 = 'AIT HUND-red'). - Millares redondos: 'TOU-SAND' (ej. 5.000 = 'FIFE TOU-SAND'). - Combinación: 38.300 = 'TREE AIT TOU-SAND TREE HUND-red'. 2. Niveles de vuelo (FL): SIEMPRE dígito a dígito (ej. FL 100 = 'FLIGHT LEVEL WUN ZE-RO ZE-RO'). 3. Escala de legibilidad: Del 1 (ininteligible) al 5 (perfecto). 'Read you 3' significa 'legible con dificultad'.

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the theoretical VHF radio communication range between an aircraft at FL 250 (25,000 ft) and a ground station at 400 ft AMSL?

- [A]: 123 NM
- [B]: 195 NM
- [C]: 250 NM
- [D]: 320 NM

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Formula: $\text{Range} = 1.23 \times \sqrt{h} = 1.23 \times \sqrt{25,000} = 1.23 \times 158.11 = 194.5 \text{ NM}$ (~195 NM).

Q2: Why must a higher frequency band be selected for HF communications during the daytime compared to night?

- [A]: Atmospheric pressure is higher during the day
- [B]: The D layer of the ionosphere forms during daytime and strongly absorbs lower HF frequencies
- [C]: Solar wind turns off the F layer
- [D]: Transmitters require cooling at night

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Solar UV radiation ionizes the lower D layer during the day, which heavily absorbs low frequencies; higher frequencies are required to penetrate to the reflecting F layer.

Q3: What is the 'Dead Space' (Skip Zone) in HF radio communications?

- [A]: An area where aircraft engines fail
- [B]: The geographic area between the limit of the ground wave and the first returning sky wave where no signal is received
- [C]: The blind cone directly above a VOR station
- [D]: Space above the ionosphere

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. The skip zone is the silent gap between the ground wave coverage limit and the touchdown point of the first ionospheric skywave.

Q4: What is the primary operational function of the SELCAL (Selective Calling) system?

- [A]: Automatically transmitting emergency MAYDAY messages
- [B]: Relieving the flight crew from continuously monitoring noisy HF audio frequencies
- [C]: Calculating aircraft ground speed
- [D]: Encoding digital transponder codes

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. SELCAL monitors the HF frequency and alerts the crew with a chime and visual annunciator only when the ground station calls their specific 4-letter code.

Q5: In which frequency band does the international aeronautical emergency frequency 121.500 MHz operate?

- [A]: HF (High Frequency)
- [B]: VHF (Very High Frequency)
- [C]: UHF (Ultra High Frequency)
- [D]: LF (Low Frequency)

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. 121.500 MHz is located in the VHF band (30 to 300 MHz).

Q6: What is the primary benefit of introducing 8.33 kHz channel spacing in European VHF communications?

- [A]: Doubling transmitter power
- [B]: Tripling the number of available communication channels to alleviate severe spectrum congestion
- [C]: Increasing VHF range to 1,000 NM
- [D]: Eliminating thunderstorm static

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Dividing 25 kHz channels into three 8.33 kHz channels triples available frequency channels.

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Chapter 3: Aerodrome Control Communications

Syllabus Source: CAE Oxford (pp. 61-88) | Pass Target: >= 90% AviationExam

1. DEPARTURE INFORMATION & PUSHBACK / START-UP

Prior to requesting start-up or taxi, the flight crew must monitor ATIS and obtain local aerodrome information:

- ATIS Acknowledgment: 'INFORMATION CHARLIE RECEIVED, QNH 1014'.
- Start-up Request: Pilot: 'REQUEST START UP'. Controller: 'START UP APPROVED' or 'EXPECT START UP AT 35'.
- Pushback Request: Pilot: 'REQUEST PUSHBACK'. Controller: 'PUSHBACK APPROVED, FACING EAST'.
- Taxi Request: Pilot: 'REQUEST TAXI'. Controller: 'TAXI TO HOLDING POINT RUNWAY 27 VIA TAXIWAY ALPHA'. Readback: 'TAXI TO HOLDING POINT RUNWAY 27 VIA TAXIWAY ALPHA, IBERIA 123'.

2. HOLDING POINT PROCEDURES & RUNWAY CROSSINGS

[!] EXAM TRAP (AVIATIONEXAM) - Runway Safety & Mandatory Holding Point Readback

- A clearance to 'TAXI TO HOLDING POINT RUNWAY 27' is NOT a clearance to enter the runway! - The aircraft MUST HOLD SHORT of the yellow illuminated runway holding position markings.
- Runway Crossing: Requires explicit, distinct clearance: 'CROSS RUNWAY 09, REPORT VACATED'. The readback MUST include the runway designator: 'CROSS RUNWAY 09, WILCO, IBERIA 123'.

3. CONDITIONAL CLEARANCES (STRICT ICAO FORMAT)

Conditional clearances involve an aircraft being cleared to line up, cross, or take off subject to a preceding movement. To eliminate runway incursions, ICAO mandates a strict 3-part phrasing sequence:

Conditional Element	Operational Purpose	Exact ICAO Phraseology Example
1. The Condition	Preceding traffic identification	'IBERIA 456, BEHIND LANDING AIRBUS A320...'
2. The Clearance	Action authorized	'...LINE UP AND WAIT RUNWAY 27...'
3. Repetition of Condition	Re-affirmation of restriction	'...BEHIND.'
Mandatory Readback	Full verbatim repetition	'BEHIND LANDING AIRBUS A320, LINE UP AND WAIT RUNWAY 27 BEHIND, IBERIA 456.'

4. TAKE-OFF, ABORTED TAKE-OFF & LANDING CLEARANCES

- Line Up & Wait: 'LINE UP AND WAIT RUNWAY 27' (or 'LINE UP RUNWAY 27 AND WAIT').
- Take-off Clearance: 'CLEARED FOR TAKE-OFF RUNWAY 27, WIND 250 DEGREES 10 KNOTS'. Readback: 'CLEARED FOR TAKE-OFF RUNWAY 27, IBERIA 123'.
- Cancelling Take-off Clearance (Before Roll): Controller: 'TAKE-OFF CLEARANCE CANCELLED, HOLD POSITION'. Pilot: 'HOLDING, IBERIA 123'.

- Aborting Take-off (Emergency During Roll): Controller: 'STOP IMMEDIATELY, IBERIA 123 STOP IMMEDIATELY'. Pilot: 'STOPPING, IBERIA 123'.
- Landing Clearance: 'CLEARED TO LAND RUNWAY 27, WIND 260 DEGREES 12 KNOTS'.
- Missed Approach / Go-Around: Controller: 'GO AROUND, CLIMB TO 3000 FEET'. Pilot: 'GOING AROUND, IBERIA 123'.
- Runway Vacated: Pilot: 'RUNWAY VACATED' (transmitted only after the entire aircraft is clear of the runway holding markings).

5. VFR TRAFFIC CIRCUIT POSITION REPORTS

Circuit Position	Flight Path Geometry	Standard ICAO Radio Phraseology
Downwind	Opposite direction to landing	Transmit aircraft type, intentions (touch-and-go / full stop): 'G-ABCD, DOWNWIND RUNWAY 27, TOUCH AND GO'.
Base Leg	Turn onto base	Transmit: 'G-ABCD, BASE RUNWAY 27'.
Final	Aligned with runway (< 4 NM)	Transmit: 'G-ABCD, FINAL RUNWAY 27'.
Long Final	Aligned at greater distance	Transmit: 'G-ABCD, LONG FINAL RUNWAY 27' (between 4 and 8 NM from touchdown).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Algoritmo Paso a Paso (Guía para Dummies): Number Pronunciation & Reading

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6. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the correct phraseology used by a controller to instruct an aircraft to abort its take-off roll immediately?

- [A]: 'Abort take-off, cancel clearance'
- [B]: 'Stop immediately, Speedbird 123 stop immediately'
- [C]: 'Speedbird 123, hold position'
- [D]: 'Discontinue take-off now'

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. The standard emergency phraseology to abort an active take-off roll is 'STOP IMMEDIATELY' repeated twice, addressed with aircraft callsign.

Q2: In a conditional clearance, what must be transmitted first by the controller?

- [A]: The clearance to line up
- [B]: The condition (identification of preceding traffic)
- [C]: The surface wind
- [D]: The aircraft callsign followed by 'cleared'

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Under ICAO standards, the condition must always precede the clearance (e.g. 'Behind landing B737, line up and wait behind').

Q3: When should a pilot transmit the phrase 'RUNWAY VACATED'?

- [A]: As soon as the main wheels touch down
- [B]: When the nose gear crosses the runway edge line
- [C]: Only when the entire aircraft has passed beyond the runway holding position marking
- [D]: When parked on the apron

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [C]. 'Runway vacated' is transmitted only when all parts of the aircraft are completely past the runway holding line.

Q4: An aircraft is instructed: 'Taxi to holding point runway 09 via taxiway Bravo'. Does this authorize the aircraft to cross the runway?

- [A]: Yes, if no traffic is visible
- [B]: No, the pilot must hold short of runway 09 until receiving an explicit clearance to cross
- [C]: Yes, after transmitting 'Crossing'
- [D]: Only in daytime VMC

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Taxi clearances to a holding point strictly terminate at the holding point. Crossing any runway requires an explicit separate clearance.

Q5: What is the correct phraseology for an aircraft commencing a missed approach on final approach?

- [A]: 'Aborting landing'
- [B]: 'Going around'
- [C]: 'Pulling up'
- [D]: 'Cancelling landing'

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. 'GOING AROUND' is the standard ICAO radiotelephony phraseology for initiating a missed approach.

Q6: At what distance from the runway threshold is an aircraft considered to be on 'Long Final'?

- [A]: Between 1 and 2 NM
- [B]: Between 4 and 8 NM
- [C]: Beyond 15 NM
- [D]: Under 1 NM

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. 'Long Final' is reported when an aircraft is turning onto final approach or tracking final at a distance greater than 4 NM (typically 4 to 8 NM).

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Chapter 4: Approach & En-route Communications

Syllabus Source: CAE Oxford (pp. 89-118) | Pass Target: >= 90% AviationExam

1. IFR DEPARTURE CLEARANCES & LEVEL INSTRUCTIONS

Initial IFR clearances contain route of flight, assigned SID, initial altitude/FL, and transponder squawk code:

- Clearance Delivery: 'IBERIA 123, CLEARED TO PARIS CHARLES DE GAULLE VIA PINUS 1A DEPARTURE, CLIMB FLIGHT LEVEL 80, SQUAWK 4215'.
- Climb & Descent Phrasing: 'CLIMB TO FLIGHT LEVEL 240', 'DESCEND TO 4000 FEET QNH 1018'.
- Transition Altitude & Level: Transition Altitude (TA) is referenced to QNH (feet). Transition Level (TL) is referenced to standard pressure 1013.25 hPa (Flight Levels). Transition Layer is the airspace between TA and TL.

2. STANDARD EN-ROUTE POSITION REPORTING

Unless operating under radar coverage where 'Radar Contact' has been confirmed and position reports waived, an IFR flight must transmit standard position reports over designated compulsory reporting points:

Report Element	Description	Standard Radiotelephony Phrasing
1. Aircraft Identification	Callsign	'IBERIA 123'
2. Position	Compulsory reporting fix	'OVER CLACTON'
3. Time	Minutes past the hour	'AT 25'
4. Flight Level / Altitude	Present level	'FLIGHT LEVEL 280'
5. Next Position & Estimate	Next fix and estimated time	'ESTIMATING OTFORD AT 42'
6. Ensuing Significant Point	Subsequent fix name	'NEXT LYDD'

3. RADAR VECTORING & SPEED CONTROL COMMANDS

- Vectoring Instructions: 'TURN LEFT HEADING 090', 'LEAVE HEADING 270 FLY HEADING 310', 'FLY HEADING 040 FOR INTERCEPT'.
- Speed Control: 'MAINTAIN 250 KNOTS', 'REDUCE TO MINIMUM CLEAN SPEED', 'RESUME NORMAL SPEED'.
- Intercepting Localizer: 'CLEARED ILS APPROACH RUNWAY 24, REPORT ESTABLISHED ON LOCALIZER'.

4. SURVEILLANCE RADAR APPROACH (SRA)

A Surveillance Radar Approach (SRA) is an instrument approach guided by a radar controller transmitting heading instructions and advisory altitudes per nautical mile from touchdown:

[!] EXAM TRAP (AVIATIONEXAM) - SRA Transmission Intervals & Missed Approach Criteria (Exam Trap)

- Transmission Frequency on Final Approach: The radar controller must transmit to the aircraft at intervals of NOT MORE THAN 5 SECONDS during final approach! - Missed Approach Rule: If no transmission is received from the controller for an interval of 5 SECONDS (or 15 seconds in some national procedures), the pilot MUST IMMEDIATELY EXECUTE A MISSED APPROACH! - Termination of SRA: The controller terminates the approach at 2 NM from touchdown (or 1/2 NM for high-precision SRA), or when the pilot reports the runway in sight.

5. CONTROLLER-PILOT DATA LINK COMMUNICATIONS (CPDLC)

CPDLC allows direct two-way digital text messaging between ground ATC automation and the aircraft FMS. Benefits include eliminating voice frequency congestion, accents, and garbled transmissions. Standard crew responses to CPDLC uplink clearances are: WILCO, UNABLE, STANDBY, or ROGER.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Algoritmo Paso a Paso (Guía para Dummies): Number Pronunciation & Read**

REGLAS MNEMOTÉCNICAS DE EXAMEN: 1. Pronunciación de números enteros: - Centenas redondas: 'HUND-red' (ej. 800 = 'AIT HUND-red'). - Millares redondos: 'TOU-SAND' (ej. 5.000 = 'FIFE TOU-SAND'). - Combinación: 38.300 = 'TREE AIT TOU-SAND TREE HUND-red'. 2. Niveles de vuelo (FL): SIEMPRE dígito a dígito (ej. FL 100 = 'FLIGHT LEVEL WUN ZE-RO ZE-RO'). 3. Escala de legibilidad: Del 1 (ininteligible) al 5 (perfecto). 'Read you 3' significa 'legible con dificultad'.

6. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What are the required elements of a standard ICAO position report in that exact sequence?**

- [A]: Callsign, Altitude, Speed, Fuel, Next point
- [B]: Callsign, Position, Time, Flight Level, Next position and estimated time, Ensuing significant point
- [C]: Callsign, Heading, Wind, Estimated time of arrival
- [D]: Callsign, Squawk code, Distance to go

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. The mandatory sequence is: Callsign -> Position -> Time -> Flight Level -> Next position & estimate -> Ensuing point.

Q2: During a Surveillance Radar Approach (SRA), what action must the pilot take if radio contact with the radar controller is lost?

- [A]: Continue descent to MDA and land visual
- [B]: Immediately execute a missed approach
- [C]: Switch to tower frequency and land
- [D]: Orbit in place

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. On SRA final approach, communications failure (> 5s silence) mandates an immediate missed approach climb.

Q3: At what approximate distance from touchdown is a standard Surveillance Radar Approach (SRA) terminated?

- [A]: 10 NM
- [B]: 5 NM
- [C]: 2 NM (or 1/2 NM if high-precision)
- [D]: At the runway threshold

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [C]. Standard SRA terminates at 2 NM from touchdown (or 0.5 NM if certified to lower minima).

Q4: What phrase does a pilot transmit upon becoming established on the ILS localizer course?

- [A]: 'Localizer captured'
- [B]: 'Established on localizer' (or 'Established runway 27')
- [C]: 'On the beam'
- [D]: 'ILS locked'

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. 'ESTABLISHED' or 'ESTABLISHED ON LOCALIZER' is the standard ICAO radiotelephony report.

Q5: What does the instruction 'RESUME NORMAL SPEED' signify to a flight crew following ATC speed restrictions?

- [A]: Accelerate to maximum speed V_{mo}
- [B]: The previously issued speed restriction is cancelled; pilot flies normal operational flight plan speeds
- [C]: Fly exactly 250 kt
- [D]: Slow down to approach speed

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. 'Resume normal speed' cancels ATC-imposed speed control, returning aircraft to operator's flight plan profile.

Q6: What is the primary operational objective of CPDLC in oceanic and en-route airspace?

- [A]: Replacing visual scans
- [B]: Eliminating voice frequency congestion and audio language misunderstandings through digital text messaging
- [C]: Automatically controlling aircraft engines
- [D]: Providing weather radar pictures

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. CPDLC replaces voice communications with structured data messages, reducing HF/VHF frequency congestion.

ATPL EASA STUDY GUIDE | 090 - COMMUNICATIONS

Chapter 5: Weather Broadcasts & Met Information

Syllabus Source: CAE Oxford (pp. 119-142) | Pass Target: >= 90% AviationExam

1. AUTOMATIC TERMINAL INFORMATION SERVICE (ATIS)

ATIS is the continuous broadcast of recorded non-control aeronautical and meteorological information at high-density aerodromes. Its primary operational purpose is to relieve ATC frequency congestion:

- Identification Designator: Each ATIS broadcast is identified by a phonetic letter from the ICAO alphabet (ALPHA through ZULU). When updated, the designator increments sequentially.
- Crew Acknowledgment: On initial contact with Approach or Tower, flight crew must state receipt of the broadcast code letter and QNH: 'INFORMATION ECHO RECEIVED, QNH 1013'.
- Voice vs D-ATIS: Voice ATIS is broadcast on dedicated VHF channels or VOR voice subcarriers. D-ATIS (Data link ATIS) transmits digital text directly into the aircraft cockpit FMS/ACARS printer.

2. STANDARD ATIS CONTENT & MESSAGE SEQUENCE

Message Element	Exact Broadcast Format Example
1. Aerodrome & Code	'GATWICK INFORMATION FOXTROT'
2. Observation Time	'TIME 1420 UTC'
3. Expected Approach & Runway	'EXPECT ILS APPROACH RUNWAY 26L'
4. Runway Surface Conditions	'RUNWAY WET WET WET'
5. Transition Level	'TRANSITION LEVEL 60'
6. Surface Wind	'WIND 240 DEGREES 15 KNOTS GUSTING 25 KNOTS'
7. Visibility & RVR	'VISIBILITY 4000 METRES, RVR RUNWAY 26L 1200 METRES'
8. Present Weather & Clouds	'MODERATE RAIN, FEW 800 FEET, BROKEN 1500 FEET'
9. Temperature, Dewpoint & QNH	'TEMPERATURE 12, DEWPOINT 10, QNH 1014'
10. Operational Warnings	'WINDSHEAR REPORTED ON FINAL APPROACH'

3. VOLMET, SIGMET & AIRMET BROADCASTS

- VOLMET: Continuous broadcast of routine aerodrome weather reports (METAR), special reports (SPECI), and 9-hour trend forecasts (TAF) for major airports, transmitted over scheduled VHF and HF frequencies. - SIGMET: Warning issued by a Meteorological Watch Office (MWO) concerning en-route weather phenomena which may affect the SAFETY OF ALL AIRCRAFT (e.g. severe turbulence, severe icing, thunderstorm lines, volcanic ash). - AIRMET: Information for low-level flights (below FL 100 or FL 150) regarding weather phenomena not included in regional forecasts (moderate icing, moderate turbulence, widespread IMC).

4. RADIOTELEPHONY WEATHER TERMINOLOGY & CAVOK

[!] EXAM TRAP (AVIATIONEXAM) - The Official Definition of 'CAVOK' (Ceiling And Visibility OK)

The term 'CAVOK' replaces visibility, present weather, and cloud groups when ALL FOUR conditions are met: 1. Visibility is 10 KILOMETRES OR MORE; 2. No cloud of operational significance (NO CLOUD BELOW 5,000 FT or below the highest Minimum Sector Altitude, whichever is greater); 3. NO CUMULONIMBUS (CB) and NO TOWERING CUMULUS (TCU) at any altitude; 4. No significant weather phenomena (no rain, snow, fog, thunderstorm, etc.).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Algoritmo Paso a Paso (Guía para Dummies): Number Pronunciation & Reading**

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5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: Which of the following conditions is mandatory to report 'CAVOK' in an aeronautical weather broadcast?**

- [A]: Visibility 5 km and sky clear
- [B]: Visibility 10 km or more, no clouds below 5,000 ft or MSA, no CB/TCU at any level, and no significant weather
- [C]: Wind calm and temperature above freezing
- [D]: QNH exactly 1013.25 hPa

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. CAVOK requires ≥ 10 km visibility, no clouds below 5,000 ft or MSA, no CB or TCU at any altitude, and no significant weather.

Q2: How does an ATIS broadcast identify that new or updated meteorological information has been published?

- [A]: A tone sounds on the emergency frequency
- [B]: The phonetic letter designator advances to the next sequential letter (e.g. from Bravo to Charlie)
- [C]: The frequency changes
- [D]: The broadcast announces 'Flash update'

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. ATIS messages are identified sequentially by phonetic letters (Alpha through Zulu) with each newly issued observation.

Q3: What type of meteorological broadcast provides warnings of severe en-route weather hazardous to ALL aircraft?

- [A]: METAR
- [B]: TAF
- [C]: SIGMET
- [D]: ATIS

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [C]. SIGMET is an en-route safety warning issued for phenomena that affect the safety of all aircraft (severe icing/turbulence, volcanic ash).

Q4: What is the primary purpose of a VOLMET broadcast?

- [A]: Providing aerodrome control tower instructions
- [B]: Providing current METARs, SPECIs, and TAFs for major international aerodromes to aircraft in flight
- [C]: Monitoring ELT distress beacons
- [D]: Broadcasting satellite navigation corrections

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. VOLMET broadcasts surface weather reports and aerodrome forecasts to airborne aircraft over VHF and HF.

Q5: How should a pilot acknowledge receipt of ATIS information 'Golf' on initial radio contact with Approach?

- [A]: 'We have the weather'
- [B]: 'Information Golf received, QNH 1018'
- [C]: 'Roger Golf'
- [D]: 'Weather checked'

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. The pilot must explicitly state the phonetic designator ('Information Golf received') and read back the altimeter setting (QNH).

Q6: In an RVR report, what does the designation 'R26L/P1500' signify?

- [A]: Runway 26 Left RVR is exactly 1,500 feet
- [B]: Runway 26 Left RVR is greater than 1,500 metres
- [C]: Runway 26 Left is closed for 1,500 seconds
- [D]: Precision approach required

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. The prefix 'P' indicates that the Runway Visual Range exceeds the maximum reporting capability of the transmissometer (greater than 1,500 m).

Chapter 6: Distress & Urgency Procedures

Syllabus Source: CAE Oxford (pp. 143-168) | Pass Target: >= 90% AviationExam

1. DISTRESS VS URGENCY STATES (ICAO DEFINITIONS)

Emergency State	Radiotelephony Callout	ICAO Definition & Aeronautical Scenarios
Distress (MAYDAY)	Spoken 3 TIMES: 'MAYDAY, MAYDAY, MAYDAY'	Condition of being threatened by grave and/or imminent danger and of requiring immediate assistance (e.g. dual engine failure, uncontrolled in-flight fire, structural failure, forced landing / ditching).
Urgency (PAN PAN)	Spoken 3 TIMES: 'PAN PAN, PAN PAN, PAN PAN'	Condition concerning the safety of an aircraft or other vehicle, or of some person on board or within sight, but which does NOT require immediate assistance (e.g. critical passenger medical condition, lost orientation with safe fuel, single engine failure in twin cruise).

2. INTERNATIONAL EMERGENCY FREQUENCIES & SQUAWK CODES

A distress or urgency call should normally be made on the frequency in current use. If unable, or if no contact is established, transmit on the international emergency frequencies:

- 121.500 MHz: International VHF Aeronautical Emergency Frequency (VHF Guard). Continuously monitored by military, civil ATC, and oceanic stations.
- 243.000 MHz: International UHF Military Emergency Frequency (UHF Guard).
- 406.0 MHz: Digital Emergency Locator Transmitter (ELT) satellite frequency. Detected by the Cospas-Sarsat constellation; transmits unique 24-bit aircraft hex code. 121.5 MHz is retained for localized homing.

[!] EXAM TRAP (AVIATIONEXAM) - The Three Master Emergency Transponder Squawks (Memorize 100%)

- Squawk 7700: GENERAL EMERGENCY (Distress or Urgency condition); - Squawk 7600: RADIO COMMUNICATION FAILURE (Complete loss of two-way radio); - Squawk 7500: UNLAWFUL INTERFERENCE (Hijacking / aircraft seizure). Note: In an emergency, select 7700 immediately on the Mode A/C transponder. This triggers automated audio and visual alarms on ATC radar consoles!

3. STRUCTURE OF A DISTRESS MESSAGE (ORDER OF PRIORITY)

Under ICAO Annex 10, a distress message must contain the following information in sequence:

Transmission Order	Exact ICAO Radiotelephony Phrasing
1. Distress Call	'MAYDAY, MAYDAY, MAYDAY'
2. Addressed Station	'MADRID RADAR' (or 'ALL STATIONS')
3. Aircraft Callsign	'IBERIA 456' (spoken 3 times on initial call)

4. Nature of Distress	'ENGINE FIRE ON ENGINE NUMBER ONE'
5. Pilot Intentions	'DESCENDING IMMEDIATELY, DIVERTING TO BARCELONA'
6. Present Position & Level	'POSITION 20 NM SOUTH OF REUS, FLIGHT LEVEL 150, HEADING 040'
7. Additional Information	'142 PERSONS ON BOARD, FUEL ENDURANCE TWO HOURS'

4. RADIO SILENCE, RELAY & EMERGENCY TERMINATION

- Imposing Radio Silence: The aircraft in distress or the controlling station can order all other traffic to cease transmitting: 'STOP TRANSMITTING, MAYDAY' (or 'SILENCE MAYDAY').
- MAYDAY RELAY: Transmitted when an aircraft observes another aircraft in distress that cannot transmit its own message: 'MAYDAY RELAY, MAYDAY RELAY, MAYDAY RELAY'.
- Termination of Distress: When the emergency has ended: Aircraft transmits 'CANCEL DISTRESS'; the controlling station transmits 'DISTRESS TRAFFIC ENDED'.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Algoritmo Paso a Paso (Guía para Dummies): Number Pronunciation & Reading

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5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: How is a condition of 'Distress' distinguished from a condition of 'Urgency' in ICAO radiotelephony?

- [A]: Distress uses PAN PAN; Urgency uses MAYDAY
- [B]: Distress indicates grave and imminent danger requiring immediate assistance; Urgency concerns safety but does not require immediate assistance
- [C]: Distress applies only on the ground
- [D]: Urgency requires immediate ditching

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Distress (MAYDAY) = grave and imminent danger requiring immediate assistance. Urgency (PAN PAN) = safety concern not requiring immediate assistance.

Q2: What is the primary international aeronautical emergency frequency in the VHF band?

- [A]: 118.000 MHz
- [B]: 121.500 MHz
- [C]: 123.450 MHz
- [D]: 243.000 MHz

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. 121.500 MHz is the international VHF aeronautical emergency frequency (VHF Guard). 243.000 MHz is UHF military guard.

Q3: Which SSR transponder code must be set immediately when an aircraft encounters an in-flight emergency

- [A]: 7000
- [B]: 7500
- [C]: 7600
- [D]: 7700

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [D]. Squawk 7700 indicates general emergency (distress or urgency). 7600 is radio failure; 7500 is unlawful interference.

Q4: What phraseology is used by an air traffic controller to impose radio silence on a frequency during an active emergency?

- [A]: 'Silence on the frequency'
- [B]: 'Stop transmitting, MAYDAY'
- [C]: 'All aircraft maintain radio silence'
- [D]: 'Standby all stations'

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Standard ICAO phraseology to impose radio silence is 'STOP TRANSMITTING, MAYDAY'.

Q5: What is the correct phraseology transmitted by an aircraft to cancel a previously declared PAN PAN urgency?

- [A]: 'Emergency cancelled'
- [B]: 'Cancel PAN PAN, situation resolved'
- [C]: 'Disregard PAN PAN'
- [D]: 'Back to normal'

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. The pilot transmits 'CANCEL PAN PAN' followed by callsign and current status.

Q6: On what satellite frequency do modern digital Emergency Locator Transmitters (ELTs) transmit encoded digital distress data?

- [A]: 121.5 MHz
- [B]: 243.0 MHz
- [C]: 406.0 MHz
- [D]: 1090 MHz

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [C]. Modern digital ELTs transmit digital distress data on 406.0 MHz to rescue satellites, retaining 121.5 MHz for final acoustic homing.

ATPL EASA STUDY GUIDE | 090 - COMMUNICATIONS

Chapter 7: Communications Failure & Light Signals

Syllabus Source: CAE Oxford (pp. 169-194) | Pass Target: >= 90% AviationExam

1. ACTION ON SUSPECTED COMMUNICATIONS FAILURE

When two-way radio communication is lost, the flight crew must perform systematic cockpit checks: 1. Check frequency selector, audio control panel switches, volume, squelch, and headset plugs; 2. Attempt contact on previous ATC frequency, company frequency, or 121.500 MHz; 3. If receiver failure is suspected, transmit blind: 'TRANSMITTING BLIND DUE TO RECEIVER FAILURE' (transmit the message twice, including present position, intentions, and time of next transmission); 4. Set transponder Mode A/C code to 7600!

2. RADIO FAILURE PROCEDURES: VMC VS IMC (ICAO ANNEX 2)

The mandatory operational procedure depends strictly on current meteorological flight conditions:

Meteorological Condition	Mandatory ICAO Operational Failure Procedure
Flight in VMC	1. Set squawk 7600; 2. Continue flight in Visual Meteorological Conditions (VMC); 3. Land at the NEAREST SUITABLE AERODROME; 4. Report arrival to ATC by the most expeditious means (telephone).
Flight in IMC (Non-Radar)	1. Set squawk 7600; 2. Maintain assigned speed and level for 20 MINUTES following failure to report over a compulsory reporting point, then conform to filed flight plan; 3. Proceed to destination nav-aid; hold until EAT (or ETA if no EAT); 4. Commence descent at EAT/ETA and land within 30 MINUTES.
Flight in IMC (Radar)	1. Set squawk 7600; 2. Maintain assigned speed and level for 7 MINUTES following 7600 selection or failure to report, then conform to flight plan; 3. Proceed to destination nav-aid, commence descent at EAT/ETA, and land within 30 MINUTES.

[!] EXAM TRAP (AVIATIONEXAM) - AviationExam Critical Numbers: 7 Minutes vs 20 Minutes in IMC

- IN AIRSPACE WHERE RADAR IS USED: Maintain assigned speed and level for 7 MINUTES, then follow flight plan! - IN AIRSPACE WHERE RADAR IS NOT USED: Maintain assigned speed and level for 20 MINUTES, then follow flight plan! - Holding & Descent: Hold at the designated destination navigation aid until the last acknowledged EAT (Expected Approach Time), or ETA if no EAT was given. Land within 30 MINUTES of EAT or ETA!

3. MASTER TABLE OF AERODROME LIGHT SIGNALS FROM CONTROL TOWER

In the event of complete radio failure, the aerodrome control tower directs aircraft using an Aldis lamp (focused optical light gun). Memorize this entire matrix for AviationExam:

Light Signal from Tower	Meaning to Aircraft IN FLIGHT	Meaning to Aircraft ON GROUND
Steady Green	CLEARED TO LAND	CLEARED FOR TAKE-OFF
Flashing Green	RETURN FOR LANDING (Wait for steady green)	CLEARED TO TAXI

Steady Red	GIVE WAY TO OTHER AIRCRAFT & CONTINUE CIRCLING	STOP
Flashing Red	AERODROME UNSAFE, DO NOT LAND	TAXI CLEAR OF RUNWAY IN USE
Flashing White	LAND AT THIS AERODROME & PROCEED TO APRON	RETURN TO STARTING POINT ON AERODROME
Red Pyrotechnic (Flare)	NOTWITHSTANDING PREVIOUS INSTRUCTIONS, DO NOT LAND FOR THE TIME BEING	DO NOT LAND FOR THE TIME BEING

4. AIRCRAFT ACKNOWLEDGEMENT OF TOWER LIGHT SIGNALS

- In Flight (Daylight): Rocking the aircraft's wings (except on final approach).
- In Flight (Night): Flashing landing lights twice, or if not equipped, flashing navigation lights twice.
- On Ground (Daylight): Moving the ailerons or rudder.
- On Ground (Night): Flashing landing lights twice, or flashing navigation lights twice.

5. SSR TRANSPONDER MODES & DATA LINK (MODE S)

Transponder Mode	Data Transmitted to ATC	Operational Role & European Mandate
Mode A	4-digit octal code (4096 combinations)	Provides identity only (e.g. Squawk 7600). Does not transmit altitude.
Mode C	Pressure altitude reporting	Transmits standard pressure altitude referenced to 1013.25 hPa in 100-foot increments.
Mode S	Discrete 24-bit aircraft address	Selective interrogation; altitude reporting in 25-foot increments; downlinks aircraft parameters (DAP: selected altitude, roll angle, IAS/Mach). Mandatory in European airspace.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

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6. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What does a STEADY RED light signal from the aerodrome control tower mean to an aircraft in flight?

- [A]: Cleared to land
- [B]: Aerodrome unsafe, do not land
- [C]: Give way to other aircraft and continue circling
- [D]: Return to starting point

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [C]. To an aircraft in flight, Steady Red means 'Give way to other aircraft and continue circling'. (On ground, it means 'Stop').

Q2: What does a FLASHING RED light signal from the control tower mean to an aircraft on the ground?

- [A]: Stop immediately
- [B]: Taxi clear of runway in use
- [C]: Cleared to taxi
- [D]: Return to starting point

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Flashing Red to an aircraft on the ground means 'Taxi clear of runway in use'. (In flight, it means 'Aerodrome unsafe, do not land').

Q3: What does a FLASHING WHITE light signal from the control tower mean to an aircraft on the ground?

- [A]: Cleared for take-off
- [B]: Stop
- [C]: Return to starting point on the aerodrome
- [D]: Cleared to taxi

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [C]. Flashing White on the ground instructs the pilot to 'Return to starting point on the aerodrome'.

Q4: An IFR aircraft experiences complete communication failure in IMC while operating in an airspace where radar is not available.

- [A]: 3 minutes
- [B]: 7 minutes
- [C]: 20 minutes
- [D]: 30 minutes

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. In radar airspace, the pilot maintains assigned speed and level for 7 minutes. In non-radar airspace, the requirement is 20 minutes.

Q5: If an aircraft experiences complete two-way communication failure in VMC, what is the mandatory course of action?

- [A]: Continue IFR flight plan to original destination
- [B]: Set transponder to 7600, continue in VMC, land at nearest suitable aerodrome, and report to ATC
- [C]: Hold over the nearest VOR for 20 minutes
- [D]: Climb to FL 150

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Under ICAO Annex 2, radio failure in VMC requires maintaining VMC, landing at the nearest suitable aerodrome, and reporting arrival.

Q6: How does an aircraft in flight acknowledge light signals from the control tower during the hours of daylight?

- [A]: By rocking the aircraft's wings (except on base and final)
- [B]: By firing a green flare
- [C]: By deploying spoilers
- [D]: By cycling the landing gear

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [A]. In daylight, an airborne aircraft acknowledges tower light signals by rocking its wings.

Q7: What does a RED PYROTECHNIC FLARE fired from the control tower signify to an aircraft on final approach?

- [A]: Cleared to land immediately
- [B]: Notwithstanding any previous instructions, do not land for the time being
- [C]: Windshear warning
- [D]: Runway surface wet

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. A red flare fired from the ground unconditionally forbids landing: 'Notwithstanding any previous instructions, do not land for the time being'.

Q8: What resolution of pressure altitude reporting is provided by Mode S transponders compared to classic Mode C?

- [A]: Exactly the same (100 ft increments)
- [B]: 25-foot increments (compared to 100 ft in Mode C)
- [C]: 1-foot increments
- [D]: 500-foot increments

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Mode S transponders report pressure altitude in fine 25-foot increments, significantly improving ACAS II and ATC conflict detection compared to Mode C's 100-foot increments.